

Physics 201 - Lecture 18

Work & Energy

→ Forces cause objects to accelerate!

↳ velocity changes

VECTORS 

SCALARS  algebra, trig, calculus

Kinetic Energy → Energy of motion.

$$K \equiv \frac{1}{2} m v^2$$

How difficult is it to stop an object that is in motion?

$$\begin{cases} v_i = \text{large} \rightarrow \text{large kinetic energy} \\ v_f = 0 \rightarrow \text{zero kinetic energy.} \end{cases}$$

$$K_i = \frac{1}{2} m v_i^2$$

$$K_f = \frac{1}{2} m v_f^2 = 0$$

$$\Delta K \equiv K_f - K_i$$

$$= 0 - \frac{1}{2} m v_i^2$$

$$\boxed{\Delta K = -\frac{1}{2} m v_i^2}$$

Stopping an object $\rightarrow$ taking kinetic energy away from

work,

energy added to
the system.

$$K = \frac{1}{2} m v^2$$

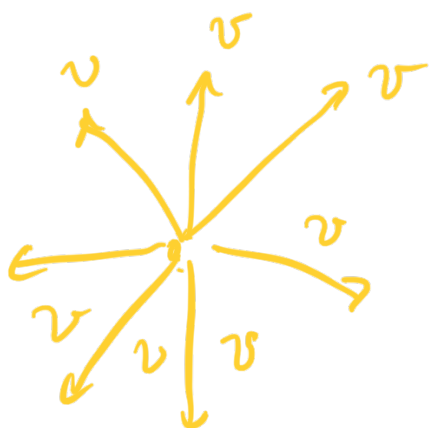
$$[K] = \text{kg} \cdot \left(\frac{\text{m}}{\text{s}}\right)^2 = \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} = \text{N} \\ = \text{Joule} \\ = \text{J}$$

How do we change the kinetic energy of an object?

→ "Work" on the object.

$$\boxed{K = \frac{1}{2} m v^2} \leftarrow \text{scalar}$$

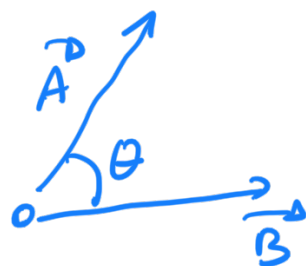
$$K = \frac{1}{2} m (\underbrace{\vec{v} \cdot \vec{v}}_{= |\vec{v}|^2 = v^2})$$



Work done by a force $\vec{F}$

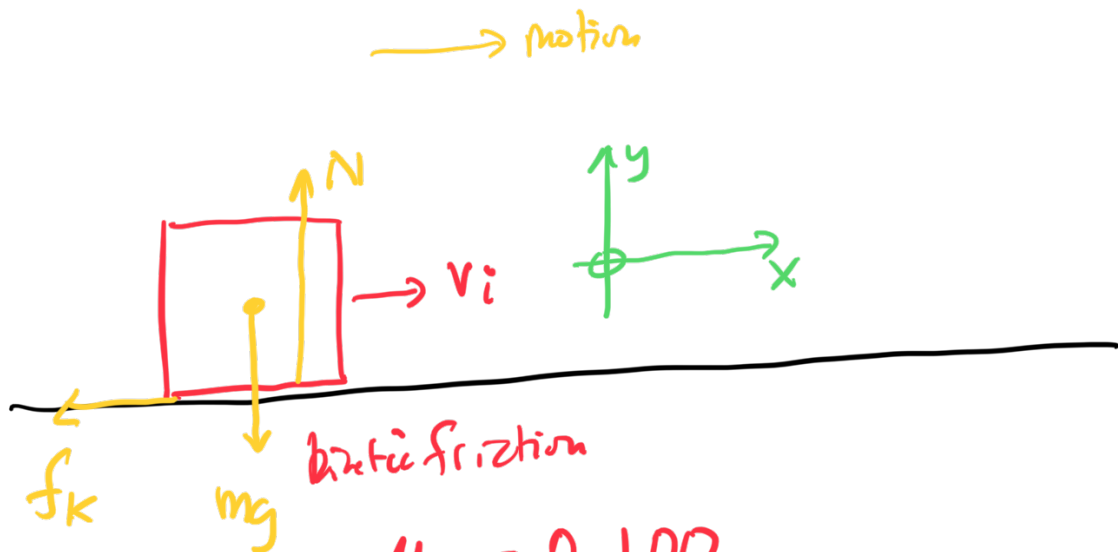
$$W_F \equiv \vec{F} \cdot \Delta \vec{x}$$

$$\left(= \int \vec{F} \cdot d\vec{x} \right)$$



$$\vec{A} \cdot \vec{B} = |\vec{A}| \cdot |\vec{B}| \cos \theta$$

$$W_F = |\vec{F}| |\Delta \vec{x}| \cos \theta$$



$$\mu_k = \underline{0.100}$$

$$m = 10 \text{ kg}$$

$$v_i = 10 \text{ m/s} \leftarrow \sim 20 \text{ mph}$$

- ① What is the acceleration?
- ② How long does it take to stop?
- ③ How far does it travel before stopping?

X | Y

$$a_x = a$$

$$\sum \vec{F}_x = m a_x$$

$$-f_k = m a_x$$

$$a_y = 0$$

$$\sum \vec{F}_y = m a_y = 0$$

$$N - mg = 0$$

$$N = mg$$

$$- \mu_k mg = m a_x$$

$$f_k = \mu_k N$$

$$= \mu_k mg$$

$$a_x = -\mu_k g$$

$$a = -0.98 \text{ m/s}^2 \quad a)$$

$$b) \quad t, \quad v_i = 10 \text{ m/s}$$

$$v_f = 0 \text{ m/s}$$

$$a = -0.98 \text{ m/s}^2$$

①

②

③

$$v_f = v_i + a t$$

$$0 = 10 - 0.98 t$$

$$10 - 10.2 \text{ s}$$

$$t = \overline{0.48} - 10.2$$

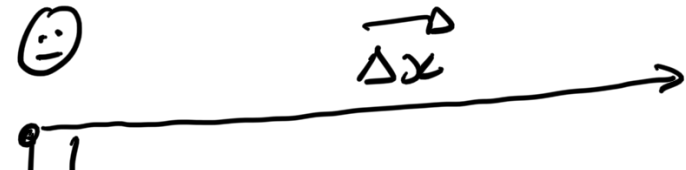
c) $\Delta x = ?$

$$\begin{aligned}\Delta x &= \cancel{v_i t} - \frac{1}{2} a t^2 \\ &= -\frac{1}{2} (-0.48) (10.2)^2 \\ &= 51.0 \text{ m}\end{aligned}$$

What is the work done by each of the forces?

$$\begin{aligned}W_{mg} &= \vec{F}_g \cdot \vec{\Delta x} \\ &= |\vec{F}_g| |\vec{\Delta x}| \cos \theta\end{aligned}$$

$\boxed{W_{mg} = 0}$ ☹️

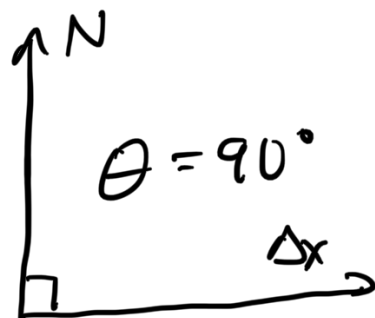


The diagram shows a horizontal line with an arrow pointing to the right, labeled $\vec{\Delta x}$. At the left end of the line is a point labeled '1'.

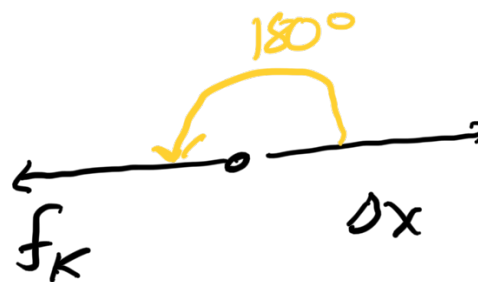


$$\cos(90^\circ) = 0!$$

$$W_N = 0$$



$$W_f = |f_k| |\Delta x| \cos \theta$$



$$W_f = -f_k \cdot \Delta x$$

$$\cos(180^\circ) = -1$$

↑ negative!

Take energy out of the system!

Work-Energy Theorem.

$$W_{\text{all forces}} = \Delta K$$

$$\cancel{W_{mg}^0} + \cancel{W_N^0} + W_f = \Delta K$$

$$W_f = \Delta K$$

$$-f_k \cdot \Delta x = \Delta K$$

$$= \cancel{\frac{1}{2} m v_f^0} - \frac{1}{2} m v_i$$

$$-f_k \cdot \Delta x = -\frac{1}{2} m v_i^2$$

$$\boxed{\cancel{f_k} \cdot \Delta x = \frac{1}{2} m v_i^2}$$

$$(\cancel{\mu_k mg})(\Delta x) = \frac{1}{2} \cancel{m} v_i^2$$

$$\Delta x = \frac{v_c^2}{2 \mu_k g}$$

$$= 51.0 \text{ m} \quad \text{😊}$$

$$\underline{\underline{W_{\text{all forces}}}} = \Delta K$$

$$\vec{F}_1 \cdot \vec{\Delta x} + \vec{F}_2 \cdot \vec{\Delta x} + \dots + \vec{F}_n \cdot \vec{\Delta x} = \Delta K$$

$$(\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \dots + \vec{F}_n) \cdot \vec{\Delta x} = \Delta K$$

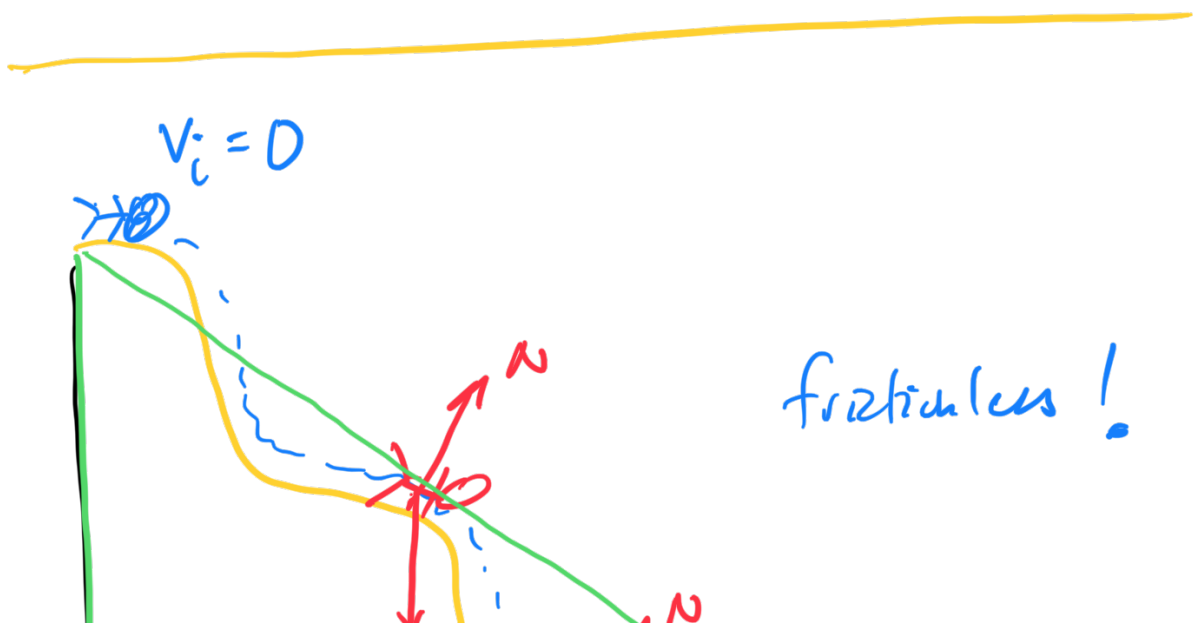
$$\underbrace{\hspace{10em}}_{\vec{F}_{\text{net}}} \cdot \vec{\Delta x} = \Delta K$$

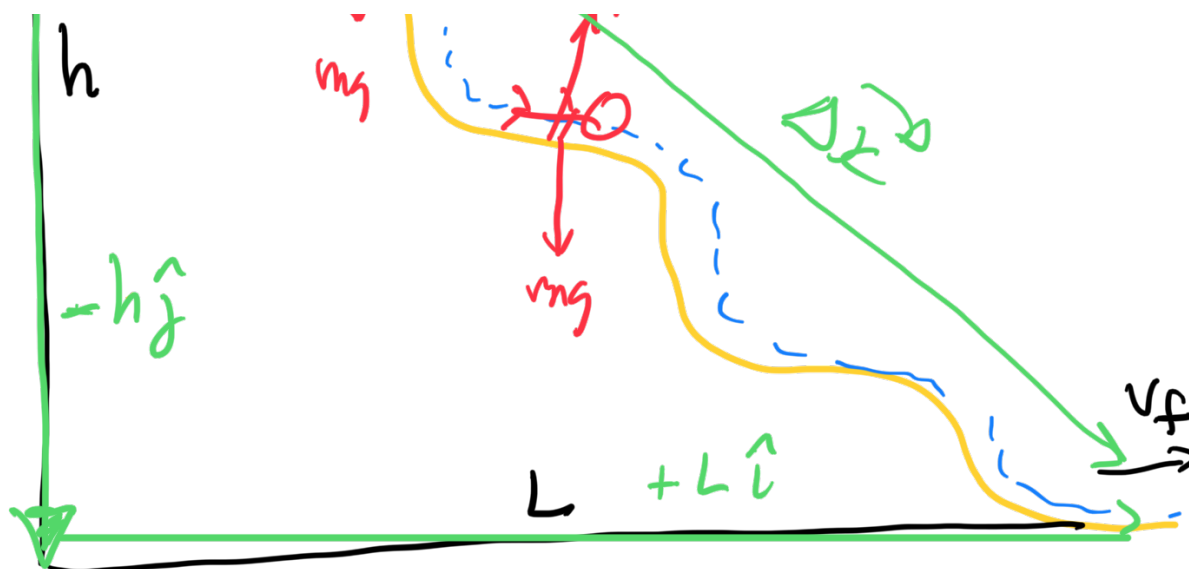
$$m \vec{a} \cdot \Delta \vec{x} = \Delta K$$

~~$$m \vec{a} \cdot \Delta \vec{x} = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$~~

$$2 \vec{a} \cdot \Delta \vec{x} = v_f^2 - v_i^2$$

$$v_f^2 = v_i^2 + 2 \vec{a} \cdot \Delta \vec{x}$$





→ gravity

→ normal force

$$W_{mg} = ?$$

$$W_N = 0$$

$$\vec{N} \perp \Delta \vec{x}$$

$$\begin{aligned}
 W_{mg} &= \underbrace{\vec{F}_g} \cdot \underbrace{\Delta \vec{x}} \\
 &= (-mg\hat{j}) \cdot (-h\hat{j} + L\hat{i}) \\
 &= mgh(\hat{j} \cdot \hat{j}) - mgL(\hat{j} \cdot \hat{i}) \\
 &= mgh(1) - mgL(0) \\
 &= mgh
 \end{aligned}$$

$$W_{mg} = mgh$$

$$\textcircled{1} \quad W_g = mgh \quad \downarrow \quad (\text{down})$$

$$\textcircled{2} \quad W_g = -mgh \quad (\text{up})$$

$$W_{\text{all forces}} = \Delta K$$

$$\cancel{W_N} + W_g = \Delta K$$

mgh

$$\cancel{mgh} = \Delta K$$

$$= \cancel{\frac{1}{2} m v_f^2} - \cancel{\frac{1}{2} m v_i^2}$$

$$v_f = \sqrt{2gh}$$

$$h = 30 \text{ m}$$

$$v_f = 24.2 \text{ m/s} \quad \sim \underline{\underline{45 \text{ mph}}}$$

